

XIAOHAN XU, 许肖汉

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RESEARCH INTERESTS

Data-Centric AI, LLM Evaluation, and Coding Agent.

EDUCATION

-  **The University of Hong Kong** *Computer Science PhD Student* Hong Kong, China 2024.9 - 2028.8
-  **University of Chinese Academy of Sciences** *IIE Cyberspace M.S.* China, 2021.9 - 2024.6
-  **Beijing Institute of Technology** *Automation B.S.* China, 2017.9 - 2021.6
-  **Illinois Institute of Technology, IIT**, *Summer Research on Big Data* U.S., 2019.7 - 2019.9
-  **University of Alberta**, *School-sponsored Exchange Program* Canada, 2018.7 - 2018.8

SELECTED PUBLICATIONS (Google Scholar Citations: 1000+)

- **BIRD-INTERACT: Re-imagining Text-to-SQL Evaluation via Lens of Dynamic Interactions.** Nan HUO*, Xiaohan Xu*, Jinyang Li*, et al. [\[ICLR 2026 Oral \(1.2%\)\]](#), 670+ GitHub stars, 1k+/month on HF.
- **LiveSQLBench: A Dynamic and Contamination-Free Benchmark for Evaluating LLMs on Real-World Text-to-SQL Tasks.** Xiaohan Xu, Jinyang Li, et al. [Homepage \(WIP\)](#), 1k+/month on HF.
- **SWE-SQL: Illuminating LLM Pathways to Solve User SQL Issues in Real-World Applications.** Jinyang Li, Xiaolong Li, Ge Qu, ..., Xiaohan Xu, ..., Reynold Cheng. [\[NeurIPS 2025 Main\]](#), 1k+ GitHub stars.
- **A Survey on Knowledge Distillation of Large Language Models.** Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, et al. [\[CSUR Under Review\]](#), 1.2k+ GitHub stars.
- **Re-Reading Improves Reasoning in Large Language Models.** Xiaohan Xu, Chongyang Tao, Tao Shen, Can Xu, Hongbo Xu, Guodong Long, Jian-guang Lou. [\[EMNLP 2024\]](#).
- **Leveraging Large Language Models for NLG Evaluation: Advances and Challenges.** Zhen Li*, Xiaohan Xu*, Tao Shen, Can Xu, Jia-Chen Gu, Chongyang Tao. [\[EMNLP 2024\]](#).
- **Subgraph Neighboring Relations Infomax for Inductive Link Prediction on Knowledge Graphs.** Xiaohan Xu, Peng Zhang, Yongquan He, Chengpeng Chao, Chaoyang Yan. [Long presentation \(25%\) at \[IJCAI 2022\]](#).

RESEARCH EXPERIENCE

BIRD-INTERACT: Dynamic Human-Centric Interaction for Data Agent

Co-First Author Collab. with [Google Cloud](#) [Homepage](#) [Paper](#) [Hugging Face](#) 2025.02-2025.09

Benchmark evaluating whether Agents can handle dynamic human-centric interaction for Text-to-SQL tasks.

- Developed an **Annotator-AI Collaborative Annotation Platform** to accelerate high-quality benchmark construction, from interaction task annotation, test case generation, environment setup, to quality iteration.
- Architected a **Modular, Server-based Agent Framework** with 3 decoupled microservices: (1) *Agent Server*, (2) *Docker-containerized DB Env. Server*, and (3) *User Simulator Server* to support easy extensibility for each module, while supporting scalable, high-parallel evaluation via asynchronous inter-service communication.
- Implemented and evaluated two **Agent Scaffoldings** based on Google ADK: (1) *Conversation* mode and (2) *Agentic* mode, both under budget-aware settings to stress-test LLM agents.
- **Outcome:** Paper accepted by [ICLR 2026 Oral \(1.2%\)](#). 670+ stars on GitHub, 1k+/month downloads on Hugging Face. Requested by [Google](#), [OpenAI](#), [Anthropic](#), and [xAI](#).

LiveSQLBench: Contamination-Free, Continuously Updating Text-to-SQL Benchmark

Project Leader Collab. with [Google Cloud](#) [Homepage](#) [Hugging Face](#) 2025.02 – Ongoing

A contamination-free, continuously updated benchmark for evaluating data agents on “live” text-to-sql tasks with newly introduced DB envs, business knowledge, and BI analytics/CRUD tasks.

- Designed an **Automatic Agent Environment Scaling Pipeline**, inspired by divide-and-conquer, to extract real-world features from raw data sources (e.g. BigQuery, CSV, DBs) to reproduce verifiable data-agent environments, supporting benchmark to remain “live”.
- Developed a **Unified Evaluation Framework for Data Agents** supporting three agent paradigms: (1) *End-to-End Prompting*, (2) *Tool-Calling Agents* with predefined DB actions, and (3) *CLI-based Agents* via terminal.

- Explored **Live Context Learning** for data agents: evaluating the ability to handle the same user query across evolving contexts where solutions may become inconsistent due to changing DB states and business knowledge.
- *Outcome: 4 releases, 1k+ down./month* on HF. Requested by **Contextual AI, Google, and xAI, etc.**

Self-Evolving Data Agent via Scaling DB Environments (WIP)

Project Leader

2025.10 – Ongoing

A study on self-evolving data agent through adaptive, co-evolving database environments that progressively challenge agents in dynamic Text-to-SQL scenarios.

- Designed **DB Environment Co-Evolution** to scale DB environments in complexity and diversity while targeting agent weaknesses (e.g., schema linking), enabling progressively harder tasks as the agent improves.
- Explored **Data Agent Self-Evolution** on the scaled DB environments: leveraging progressive challenges and failure feedback to enable continuous self-improvement and generalization.
- *Preliminary Outcome:* Achieved **+25.1% improvement via SFT** ; actively extending to Agentic RL training.

Data Intelligence Index for AI

HKU BIRD Team [Homepage](#)

2026.01 – Ongoing

Comprehensive evaluation of frontier AI models on various data-centric skills.

- Built the **Data Intelligence Index**, evaluating frontier models across 6 data-centric dimensions (e.g., DB querying, BI analysis), providing systematic capability and cost-performance profiling for models.
- Trained a lightweight **Data AI Router** using a small LLM (*Qwen2.5-0.5B*) to dynamically route queries to the suitable frontier model based on task and user preference, improving overall accuracy while reducing cost.

INTERNSHIP EXPERIENCE

 **Microsoft** Research Intern 2023.05 – 2023.10

- **LLM Reasoning:** Proposed a re-reading method to enhance bidirectional reasoning for decoder-only LLMs, improving question comprehension in complex reasoning tasks (**EMNLP 2024**).
- **Multi-Agent Reasoning:** Explored collaborative reasoning by designing multiple agents with distinct reasoning strategies that communicate to improve reasoning performance.
- **LLM KD and Evaluation:** Two survey papers on *LLM knowledge distillation* and *evaluation* (**EMNLP 2024**).

 **Baidu Inc. | XiaoDu Cloud** Research Intern 2022.11 – 2023.05

- **LLM Post-Training:** Contributed to instruction alignment of LLMs via supervised fine-tuning (SFT), including building **1M+ Chinese instruction data** through self-instruct and quality filtering.
- **Distributed LLM Training:** Conducted distributed SFT training using DeepSpeed and MPI, and analyzed the impact of data quality, format, and scale on instruction-following ability.
- **Educational QA Model:** Built an instruction-tuned model for K–9 multi-subject exam solving with knowledge injection, achieving 82% accuracy on English and 61% on Chinese question sets.

 **BAAI | Recognition & Data Group** Research Intern 2022.04 – 2022.10

- **Emotional Support Dialogue:** Developed a dialogue model for emotional support by incorporating prior conversational knowledge and latent variables to improve response quality and diversity, achieving SOTA on both automatic and human evaluations.
- **PLM + Knowledge Graphs:** Explored integrating pretrained language models with knowledge graphs via contrastive learning and multi-task learning (MLM + link prediction).

AWARDS & HONORS

- **Postgraduate Scholarship (PGS) , HKU, 2024** 2024
- **Outstanding Graduates, IIE, UCAS** 2024
- **IIE President Special Award (Top 1%)** 2024
- **National Scholarship (< 2%)** 2018
- **National Encouragement Scholarship (< 5%)** 2019